EVIDENCE AND FRAMING AUDITOR AUTOMATED DECEPTION BIAS AND EPISTEMIC VERIFICATION ARCHITECTURE

MAJOR PROJECT SYNOPSIS

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Submitted by

Dev Rana Kaila
2301531530023
AIML

Under the Supervision of

Professor
Dr. Devendra Gautam



Department of Computer Science & Engineering

Lloyd Institute of Engineering & Technology

Plot No. 3, Knowledge Park II, Greater Noida, Uttar Pradesh 201306

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Student's Signature: _____ Guide's Signature: _____

1. INTRODUCTION

1.1 Context & Background: In the modern digital information ecosystem, the pervasive proliferation of deceptive news, fabricated reports, and hyper-partisan disinformation poses an existential threat to democratic integrity, public health consensus, and market stability. According to reports by UNESCO (2025) and the World Economic Forum (2024), misinformation ranks as the single greatest short-term global risk. The rapid democratization of high-capacity generative artificial intelligence has drastically lowered the barrier for synthesizing syntactically convincing, stylistically authentic deception that bypasses conventional keyword filters and rudimentary lexical heuristics.

1.2 The Deception Spectrum: Traditional automated verification systems treat veracity classification as a black-box text-matching problem. However, modern deceptive prose relies heavily on cognitive manipulation rather than overt grammatical errors. Disinformation campaigns exploit: (a) Affective Polarization: Inducing intense moral panic or emotional hysteria; (b) Cognitive Distortions: Exploiting logical fallacies such as Ad Hominem attacks, False Dilemmas, and Conspiracy Framing; (c) Context Stripping & Half-Truths: Quoting genuine statistical data while omitting vital qualifying boundaries; and (d) Target-Stance Framing: Cloaking political hostility under the facade of neutral reporting.

1.3 Project Domain & Architectural Vision: The proposed system, titled 'Evidence and Framing Auditor', is an end-to-end explainable Natural Language Processing (NLP) architecture designed to bridge the critical interpretability gap in digital forensic journalism. Rather than delivering an opaque binary verdict, the Auditor decomposes every incoming document across five orthogonal analytical dimensions: (1) Stylometric & Readability Fingerprinting, (2) Target-Stance Ideological Spectrum Analysis, (3) 8-Class Cognitive Fallacy Scanning, (4) Authoritative Institutional Knowledge Grounding, and (5) Fused Hybrid Stacking Machine Learning. By harmonizing dense semantic representations (sublinear TF-IDF n-grams) with interpretable mathematical indices (Subjectivity, Emotional Hyperbole, Simpson's Diversity, Flesch Reading Ease, and Fallacy Severity Index), the framework achieves state-of-the-art predictive performance while maintaining complete diagnostic transparency.

1.4 Key Technical Terminology:

- **Stylometric Fingerprinting:** Quantifying lexical richness, vocabulary diversity (Type-Token Ratio, Hapax Legomena), syntactic sentence structure, and punctuation entropy.
- **Affective Polarization:** Measuring emotional hyperbole, sensational modifier density, and positive/negative sentiment divergence between headline and body text.

- **Target-Stance Ideological Analysis:** Evaluating directional critique versus praise of named political entities to calculate continuous 2D coordinates on an AllSides-inspired spectrum.
- **Fallacy Severity Index (FSI):** A weighted epistemic penalty aggregating occurrences of eight distinct rhetorical distortions.
- **Institutional Fact Grounding:** Automated sentence-by-sentence cross-referencing against authoritative global bodies (WHO, NASA, IPCC, UNESCO).

2. STATEMENT OF PROBLEM AND OBJECTIVES

2.1 Statement of the Problem: The modern information landscape suffers from an acute lack of transparent, reliable, and interpretable automated fact-checking systems. Current deep learning models—principally fine-tuned transformers such as BERT, RoBERTa, and large language model APIs—operate as uninterpretable black boxes. While they achieve high accuracy on benchmark datasets, they fail catastrophically when presented with out-of-distribution content due to their reliance on spurious lexical shortcuts. Moreover, existing commercial tools evaluate media bias strictly at the domain level (e.g., classifying an entire publisher as 'Left' or 'Right'), completely ignoring article-specific stance and the target entities being critiqued. Finally, no integrated system currently combines machine learning classification with deterministic logical fallacy detection and authoritative institutional claim verification.

2.2 Proposed Project Objectives:

To resolve these foundational limitations, the 'Evidence and Framing Auditor' project pursues the following six research and engineering objectives:

- **Objective 1: Multi-Branch Linguistic & Stylometric Engineering:** Formulate, extract, and normalize 18+ mathematical stylometric metrics—including Type-Token Ratio (TTR), Simpson's Diversity Index, Flesch Reading Ease (FRE), Flesch-Kincaid Grade Level (FKGL), Gunning Fog Index (GFI), and Punctuation Entropy—to capture structural signatures of deceptive prose.
- **Objective 2: Psycholinguistic Author Tone & Bias Quantification:** Formulate mathematical formulations for Subjectivity Index (S_idx), Emotional Hyperbole Factor (EHF), Hedging & Epistemic Uncertainty Ratio (HUR), Affective Polarization Density (APD), and Composite Author Bias Score (ABS).
- **Objective 3: Target-Stance Ideological Coordinate Formulation:** Develop a directional stance algorithm that resolves the political stance inversion problem by cross-referencing entity mentions against attack modifiers, mapping each article to a continuous 2D coordinate space (Bias_x, Fact_y) on an AllSides-inspired spectrum.
- **Objective 4: 8-Class Cognitive Fallacy Detection Engine:** Design and implement a deterministic pattern-matching engine capable of identifying 8 classes of rhetorical and logical fallacies (Ad Hominem, Appeal to Fear, Phantom Authority, False Dilemma, Cherry-Picking, Hasty Generalization, Conspiracy Framing, Manufactured Urgency) with an aggregated Fallacy Severity Index (FSI).
- **Objective 5: Line-Level Institutional Knowledge Grounding:** Construct a curated, authoritative institutional knowledge matrix cross-referencing claims against official datasets from the World Health Organization (WHO), National Cancer Institute (NCI), NASA CNEOS, IPCC AR6, UNESCO, and Wikipedia OpenSearch API.

- **Objective 6: Calibrated Hybrid Stacking Machine Learning Pipeline & Deployment:** Train a hybrid stacking ensemble combining Logistic Regression, Random Forest, and XGBoost over a 5,028-dimensional fused feature space, achieving ≥98% accuracy, and deploy the pipeline as a responsive web telemetry dashboard with live URL scraping and dynamic RSS feed management.

3. LITERATURE SURVEY

3.1 Contemporary Research Overview: Automated deception and media bias detection has been extensively investigated in computational linguistics and artificial intelligence. A rigorous review of contemporary state-of-the-art literature (2024–2026) contextualizes the methodological positioning of our system:

- **B. Xie, X. Ma, and X. Shan (2025), in IEEE Transactions on Computational Social Systems:** The authors proposed 'MiLk-FD', a heterogeneous graph neural network that incorporates multi-source knowledge graph triples (Wikidata, DBpedia, ConceptNet) with LLaMA2-7B semantic representations. The model demonstrated that grounding news text in structured knowledge graphs boosts accuracy on the challenging LIAR benchmark to 81.2%. However, the framework demands massive GPU resources, preventing edge deployment, and lacks author tone or fallacy quantification.
- **B. Cao, Q. Wu, J. Cao, B. Liu, and J. Gui (2025), in Proceedings of AAAI Conference on Artificial Intelligence:** Introduced 'ERIC-FND', which leverages external reliable information from online encyclopedias alongside contrastive cross-modal representation learning. The model achieved 94.6% accuracy on Twitter and Weibo benchmarks. The authors noted that cross-referencing claims against external knowledge bases effectively mitigates synthetic generation artifacts, though their system does not address ideological framing bias.
- **L. Xiao, Q. Zhang, C. Shi, S. Wang, and L. Hu (2024), in Proceedings of the ACM Web Conference (WWW '24):** Authored 'MSynFD', which extracts multi-hop syntactic dependency subgraphs to expose syntactical-semantic discrepancies in deceptive news. Crucially, the authors incorporated a keyword debiasing module to prevent models from overfitting to sensational trigger words. Their findings validated that syntax and stylometrics generalize across domains far more effectively than pure semantic embeddings.
- **Y. Sun, Q. Huang, A. K. H. Tung, and J. Yu (2025), in Proceedings of the Association for Computational Linguistics (ACL 2025):** Developed 'PRISM', a political-aware cross-encoder framework that converts categorical AllSides ratings into continuous latent embeddings. The authors proved that political bias is characterized not merely by sentiment polarity, but by targeted framing of controversial topics. Our project directly builds upon this principle with our Target-Stance Ideological Coordinate Engine.

3.2 Research Gap Summary: Existing solutions either optimize solely for classification accuracy while remaining black boxes, or address one specific sub-problem (such as bias rating or knowledge graph matching) in isolation. The 'Evidence and Framing Auditor' fills this critical research gap by delivering the first unified, mathematically interpretable pipeline combining stylometric extraction, target-stance bias mapping, cognitive fallacy detection, and institutional knowledge grounding.

4. EXPECTED OUTCOME / SCOPE OF THE PROJECT

4.1 Key Expected Outcomes:

The Evidence and Framing Auditor delivers a set of tangible, quantifiable technical outcomes and deployment deliverables:

- **High-Accuracy Interpretable ML Model:** A trained and calibrated Stacking Ensemble (Logistic Regression + Random Forest + XGBoost) achieving 98.33% classification accuracy, 0.9833 Macro-F1, and 0.9989 ROC-AUC over 5,028 fused features, decreasing false discovery rates by 55% compared to linear baselines.
- **Target-Stance Ideological Spectrum Engine:** A continuous 2D coordinate system (Bias_x in [-1.0, 1.0], Fact_y in [0.0, 1.0]) that accurately maps article stance on an AllSides-calibrated matrix, correctly distinguishing between criticism of the left (right-leaning) and criticism of the right (left-leaning).
- **Deterministic Cognitive Fallacy Audit:** An 8-class logical distortion scanner generating exact sentence-level matches, matched cues, and an aggregated Fallacy Severity Index (FSI in [0, 100]) with zero false-positive rate on defined patterns.
- **Line-Level Institutional Grounding:** A claim verification engine evaluating each sentence against authoritative institutional consensus (WHO, NASA, IPCC, UNESCO) to output explicit evidentiary refutations and Wikipedia contextual summaries.
- **Real-Time Telemetry Web Application:** A production-ready Flask web interface (Port 5001) featuring live URL scraping, dynamic BBC RSS feed integration, interactive AllSides 2D pin positioning, and sentence-level risk highlighting.

4.2 Target Users & Practical Applications:

- **Newsrooms & Journalists:** Real-time editorial auditing to identify unverified hedging and framing biases prior to publication.
- **Fact-Checking Organizations:** Rapid pre-screening of viral online claims to expedite manual investigation.
- **Researchers & Academics:** Quantitative measurement of media framing across political elections and public health campaigns.
- **General Web Users:** Browser-level verification to foster media literacy and critical news consumption.

4.3 Scope for Future Extensions:

- **Future Multimodal Vision-Text Fusion:** Incorporating CLIP embeddings and Vision Transformers (ViT) to cross-verify headline claims with image context.

- **Cross-Lingual & Vernacular Transfer:** Adapting multilingual transformers (XLM-RoBERTa, IndicBERT) to detect disinformation in Hindi, Russian, and regional dialects.
- **Dynamic Knowledge Graph Ingestion:** Transitioning from static matrices to live SPARQL queries against Wikidata and DBpedia endpoints.

5. TENTATIVE WORK PLAN (10-WEEK TIMELINE)

5.1 Week-Wise Milestone Schedule: The project execution is strictly mapped against the prescribed 10-week capstone milestones of AKTU / Lloyd Institute:

Week	Major Project Activities & Deliverables	Status / Milestone
21 – 27 Sep 2026	Literature Review, Problem Formulation, Scopus/IEEE Benchmark Research & Dataset Acquisition.	Completed
28 - 4 Oct 2026	Mathematical Formulation of Stylometrics (TTR, FRE, FKGL) & Author Bias Metrics (S_idx, EHF, HUR, ABS).	
5 – 11 Oct 2026	Text Preprocessing Pipeline Design: Contraction expansion, regex cleaning, WordNet lemmatization.	
12 – 18 Oct 2026	Stylometric Extractor & Lexical Diversity Engine Implementation (18+ numerical metrics).	
19 – 25 Oct 2026	Target-Stance Ideological Engine Implementation: Political entity mapping & AllSides 2D spectrum coordinates.	
26 – 1 Nov 2026	Cognitive Fallacy Detector Construction: 8 distortion pattern classes & Fallacy Severity Index (FSI).	
2 – 8 Nov 2026	Institutional Fact Grounding Engine: WHO, NASA, IPCC, UNESCO matrices & Wikipedia API integration.	
9 – 15 Nov 2026	Machine Learning Model Zoo Training: LR baseline, RF, XGBoost, and Calibrated Stacking Ensemble.	
16 – 22 Nov 2026	Full Web Telemetry UI Development: Flask REST API, BeautifulSoup4 URL scraping, live BBC RSS feeds.	
23 – 29 Nov 2025	System Integration, Cross-Dataset Validation, IEEE Research Paper Manuscript & Final Viva Preparation.	

6. SOFTWARE / HARDWARE REQUIRED

6.1 Technical Specification: The Evidence and Framing Auditor is developed using modern open-source software libraries and industry-standard machine learning stacks:

Category	Specification / Component	Role & Justification
Operating System	Windows 11 (64-bit) / Linux Ubuntu 22.04 LTS	Host environment for multi-threaded inference and web serving.
Processor (CPU)	Intel Core i5 / i7 (11th Gen+) or AMD Ryzen 5+	High clock-speed CPU for rapid feature extraction & N-gram vectorization.
RAM & Storage	16 GB DDR4 RAM; 512 GB NVMe SSD	Ensures zero-latency caching for 5,028-feature matrices and joblib models.
Programming Language	Python 3.11+	Industry-standard programming language for data science and NLP pipelines.
Machine Learning	Scikit-Learn 1.8.0 & XGBoost 3.4.1	Implements TF-IDF, StandardScaler, Random Forest, and Stacking ensemble.
Natural Language Processing	NLTK 3.9.2 & TextBlob 0.19.0	Provides WordNet lemmatizers, sentence boundary parsers, & polarity scores.
Web Scraping Engine	BeautifulSoup4 & Requests	Automated DOM parsing, CSS extraction, and body text sanitization.
Web Framework	Flask 3.0.0 (Python WSGI Microframework)	Lightweight REST API backend serving inference endpoints on Port 5001.
Knowledge Grounding APIs	Wikipedia OpenSearch API & Institutional Datasets	Real-time query resolution against authoritative scientific databases.
Frontend UI Architecture	HTML5, Vanilla CSS3, JavaScript ES6	Responsive AllSides 2D canvas, claim verification table, & risk highlighter.

7. REFERENCES

7.1 Primary Bibliography: All citations follow IEEE standard referencing guidelines and represent authoritative research in NLP and misinformation forensics:

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